

# Exploratory Data Analysis

```
library(haven)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
dat25 <- read_xpt("../data/LLCP2025.XPT")
dat24 <- read_xpt("../data/LLCP2024.XPT")
dat23 <- read_xpt("../data/LLCP2023.XPT")
dat22 <- read_xpt("../data/LLCP2022.XPT")
dat21 <- read_xpt("../data/LLCP2021.XPT")
dat20 <- read_xpt("../data/LLCP2020.XPT")
dat19 <- read_xpt("../data/LLCP2019.XPT")
dat18 <- read_xpt("../data/LLCP2018.XPT")
```

## Exploratory data analysis

### variables asked only in 2025, 2024, 2022, and 2019

NOBCUSE9 = What was main reason for not doing anything to keep you from getting pregnant?

PFPPRVN4 = The last time you had sexual intercourse, did you or your partner do anything to

keep you from getting pregnant? TYPCNT10 = The last time you had sexual intercourse, what did you or your partner do to keep you from getting pregnant? What was your main reason for not doing anything to prevent pregnancy the last time you had sexual intercourse?

## variables asked in all surveys

SEXVAR = 2 if female

PREGNANT = current pregnancy status (1=yes, 2=no, 7=don't know, 9=refused)

MENTHLTH = Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? (1-30, 88=none, 77=don't know, 99=refused)

EMPLOY1 = employment status (1 = employed for wages, 2 = self-employed, 3 = out of work for 1 year or more, 4 = out of work for less than 1 year, 5 = homemaker, 6 = student, 7 = retired, 8 = unable to work, 9 = refused)

\_INCOMG1 = income categories (1=less than 15k, 2=15k to 25k, 3= 25k to 35k, 4= 35k to 50k, 5=50k to 100k, 6=100k to 200k, 7=200k or more, 9=don't know)

EDUCA = What is the highest grade or year of school you completed? (1=no high school, 2=high school, 3=some college, 4=college, 9=don't know)

\_HCVU654 = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don't know)

\_RACE = Race/ethnicity categories (1=nh white, 2=nh black, 3=nh native american, 4=nh asian, 5=nh native hawaiian, 7=nh multiracial, 9=don't know)

**subset each survey year to include relevant variables, subsetting the sample to only women**

```
sub_dat25 <- dat25 |> select(c("_STATE", "SEXVAR", "PREGNANT", "MENTHLTH",  
  ↪ "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE")) |> filter(SEXVAR==2)  
  ↪ |> select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',  
  ↪ EMPLOY='EMPLOY1')  
  
sub_dat24 <- dat24 |> select(c("_STATE", "SEXVAR", "PREGNANT", "MENTHLTH",  
  ↪ "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE")) |> filter(SEXVAR==2)  
  ↪ |> select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',  
  ↪ EMPLOY='EMPLOY1')
```

```

sub_dat23 <- dat23 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU653", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU653', EMPLOY='EMPLOY1')

# Roe v. Wade was overturned on June 24, 2022
sub_dat22 <- dat22 |>
  ↪ select(c("_STATE","SEXVAR","IMONTH","IDAY","IYEAR","PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU652", "_RACE1")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU652', EMPLOY='EMPLOY1') |> mutate(
  IYEAR = as.integer(as.character(IYEAR)),
  IMONTH = as.integer(as.character(IMONTH)),
  IDAY = as.integer(as.character(IDAY)),
  date = as.Date(sprintf("%04d-%02d-%02d", IYEAR, IMONTH, IDAY)),
  post_ban = ifelse(
    date >= as.Date("2022-06-24"),1,0)
)

sub_dat21 <- dat21 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU652", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU652', EMPLOY='EMPLOY1')

sub_dat20 <- dat20 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG',
  ↪ INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

sub_dat19 <- dat19 |>
  ↪ select(c("_STATE","SEXVAR","PREGNANT","MENTHLTH","EMPLOY1", "_INCOMG",
  ↪ "EDUCA", "_HCVU651", "_RACE")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
  ↪ rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

sub_dat18 <- dat18 |> select(c("_STATE", "SEX1","PREGNANT", "MENTHLTH",
  ↪ "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE")) |> filter(SEX1==2)
  ↪ |> select(-SEX1) |> rename(INCOME='_INCOMG', INSURANCE='_HCVU651',
  ↪ EMPLOY='EMPLOY1')

data_list <- list(
  `2018` = sub_dat18,
  `2019` = sub_dat19,

```

```

`2020` = sub_dat20,
`2021` = sub_dat21,
`2022` = sub_dat22,
`2023` = sub_dat23,
`2024` = sub_dat24,
`2025` = sub_dat25
)

vars_to_keep <- c(
  "_STATE",
  "PREGNANT",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE",
  "IMONTH",
  "IDAY",
  "IYEAR"
)

analysis_data <- purrr::imap_dfr(
  data_list,
  ~ .x |>
    select(any_of(vars_to_keep)) |>
    mutate(YEAR = as.integer(.y))
) |> mutate(
  post_ban = case_when(
    YEAR <= 2021 ~ 0,
    YEAR >= 2023 ~ 1
  )
) |> select(-c(IMONTH, IDAY, IYEAR)) |> rename(STATE = "_STATE")

analysis_data

```

# A tibble: 1,813,542 x 9

|   | STATE | PREGNANT | MENTHLTH | EMPLOY | INCOME | EDUCA | INSURANCE | YEAR  | post_ban |
|---|-------|----------|----------|--------|--------|-------|-----------|-------|----------|
|   | <dbl> | <dbl>    | <dbl>    | <dbl>  | <dbl>  | <dbl> | <dbl>     | <int> | <dbl>    |
| 1 | 1     | NA       | 88       | 2      | 4      | 6     | 9         | 2018  | 0        |
| 2 | 1     | 2        | 88       | 1      | 2      | 6     | 2         | 2018  | 0        |
| 3 | 1     | NA       | 88       | 7      | 2      | 4     | 9         | 2018  | 0        |

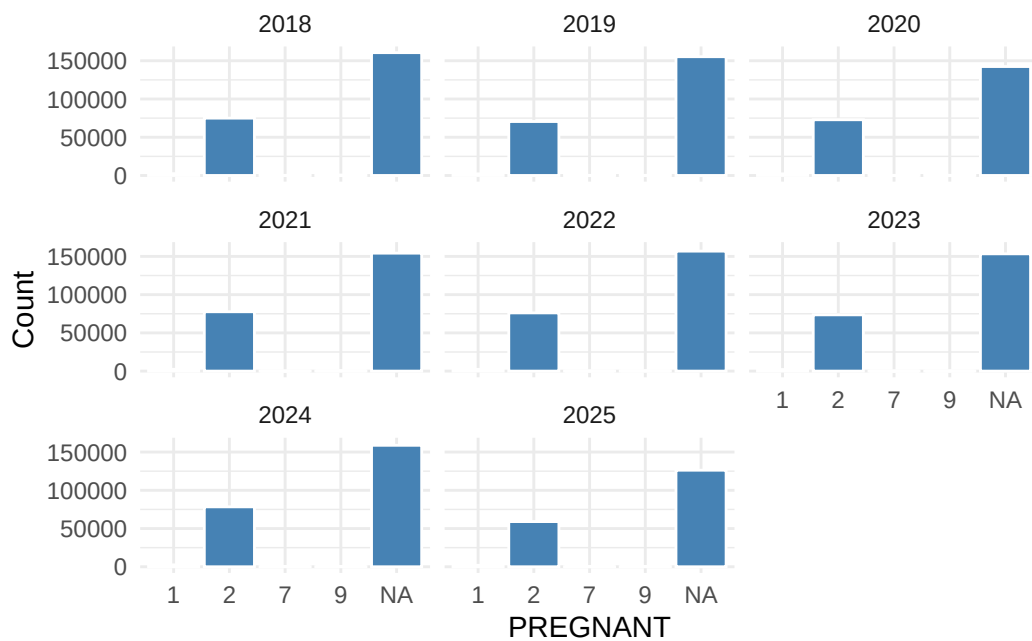
|    |   |    |    |   |   |   |   |      |   |
|----|---|----|----|---|---|---|---|------|---|
| 4  | 1 | 2  | 88 | 1 | 9 | 6 | 1 | 2018 | 0 |
| 5  | 1 | NA | 88 | 7 | 9 | 5 | 9 | 2018 | 0 |
| 6  | 1 | NA | 88 | 5 | 5 | 6 | 9 | 2018 | 0 |
| 7  | 1 | 2  | 88 | 5 | 9 | 6 | 1 | 2018 | 0 |
| 8  | 1 | 2  | 88 | 1 | 5 | 6 | 1 | 2018 | 0 |
| 9  | 1 | NA | 2  | 2 | 4 | 5 | 1 | 2018 | 0 |
| 10 | 1 | 2  | 1  | 3 | 3 | 5 | 2 | 2018 | 0 |

# i 1,813,532 more rows

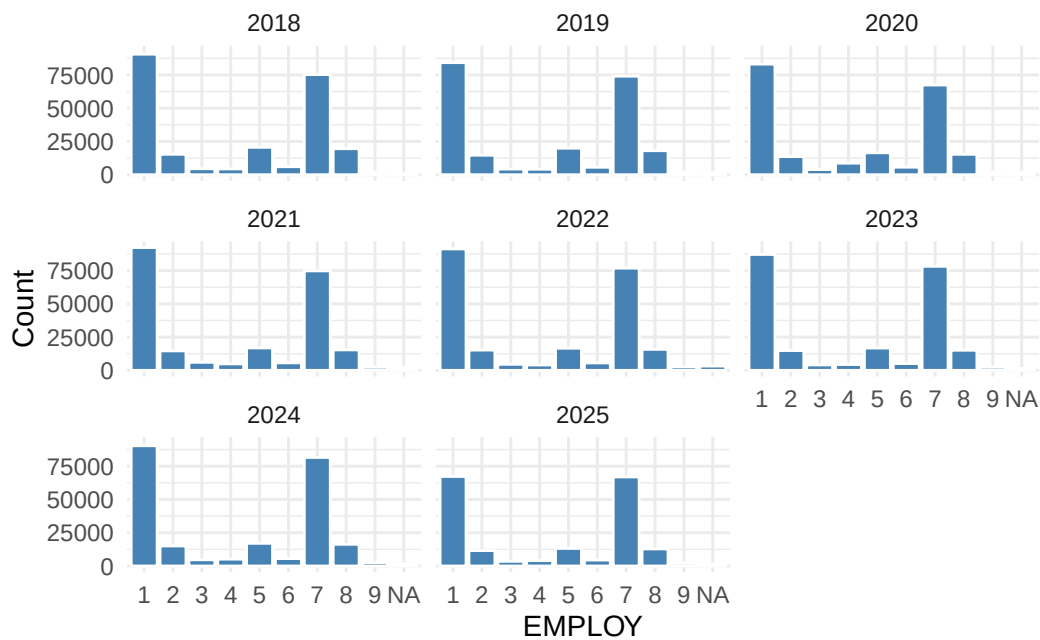
## visualizing data

```
# categorical variables
apply(names(analysis_data)[!names(analysis_data) %in%
  ↪ c("STATE","YEAR","post_ban", "MENTHLTH")], function(var){
  ggplot(
    analysis_data,
    aes(x = factor(.data[[var]]))
  ) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    x = var,
    y = "Count"
  )
})
```

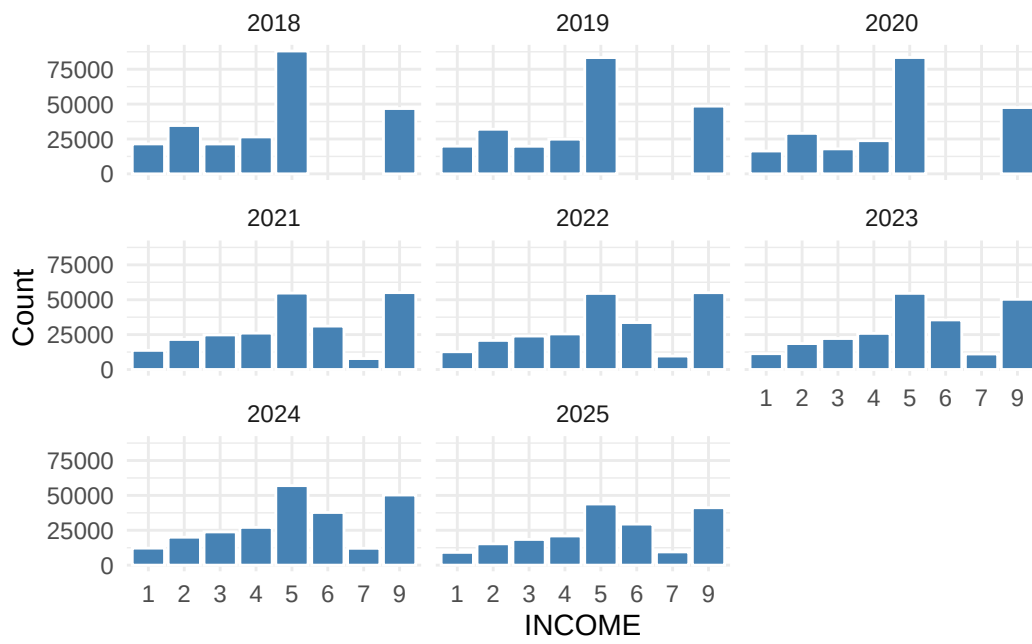
\$PREGNANT



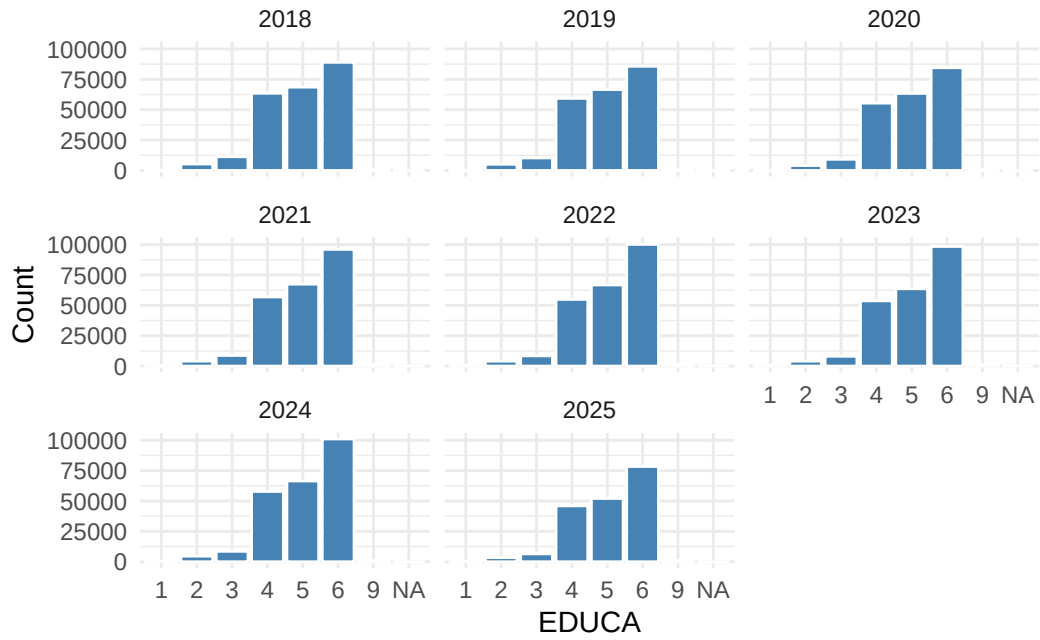
## \$EMPLOY



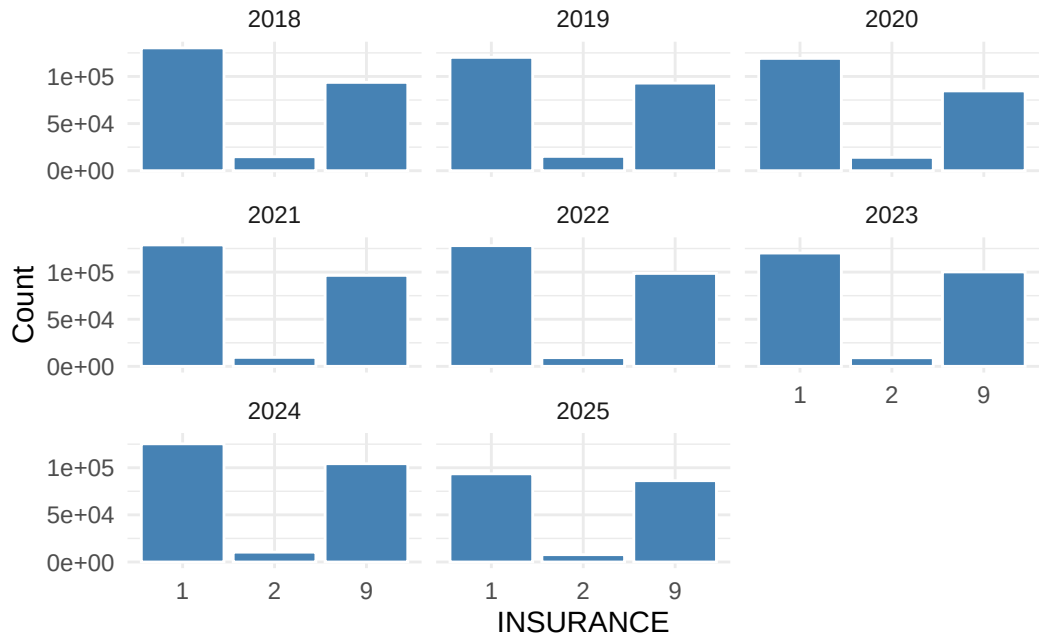
\$INCOME



\$EDUCA



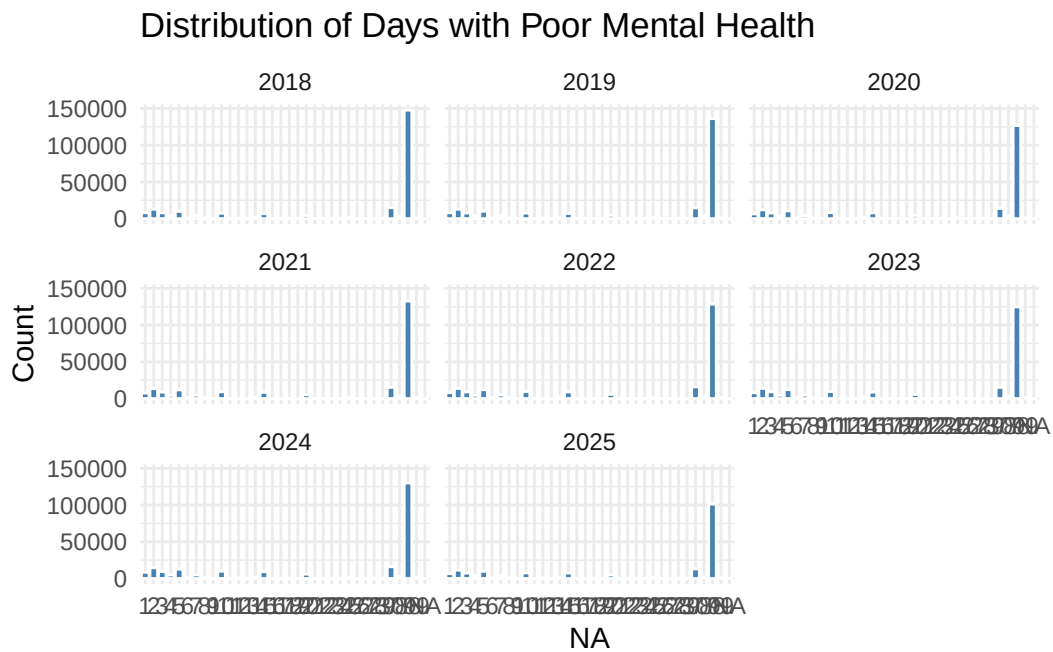
\$INSURANCE





```
# numerical variable
ggplot(
  analysis_data ,
  aes(x = factor(MENTHLTH))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    title = "Distribution of Days with Poor Mental Health",
    x = var,
    y = "Count"
  )
)
```

Warning in label(labels[[nm]] %||% ""): NAs introduced by coercion



## summary statistics

```
analysis_data |> group_by(YEAR, PREGNANT) |> summarize(n=n(),  
                                                    .groups="drop_last")  
↪ |>  
  
mutate(percent = 100 * n /sum(n)) |>  
ungroup()
```

```
# A tibble: 40 x 4  
  YEAR PREGNANT      n percent  
  <int>   <dbl> <int>   <dbl>  
1  2018         1   2398    1.00  
2  2018         2  75001   31.4  
3  2018         7    225   0.0942  
4  2018         9    660   0.276  
5  2018        NA 160627   67.2  
6  2019         1   2121   0.929  
7  2019         2  70582   30.9  
8  2019         7    180   0.0788  
9  2019         9    199   0.0871  
10 2019        NA 155337   68.0  
# i 30 more rows
```

```
get_summary_stats_over_entire_period <- function(data, var) {  
  
  tab <- data |>  
    janitor::tabyl(  
      {{ var }},  
      show_na = TRUE  
    )  
  
  # The first column is the categorical variable  
  category_col <- names(tab)[1]  
  
  total_n <- sum(tab$n)  
  
  valid_n <- sum(  
    tab$n[!is.na(tab[[category_col]])]  
  )  
  
  tab |>
```

```

dplyr::mutate(
  percent = 100 * n / total_n,
  valid_percent = dplyr::if_else(
    valid_n > 0 & !is.na(.data[[category_col]]),
    100 * n / valid_n,
    NA_real_
  )
)
}
get_summary_stats_over_entire_period(analysis_data,PREGNANT)

```

| PREGNANT | n       | percent     | valid_percent |
|----------|---------|-------------|---------------|
| 1        | 17907   | 0.98740476  | 2.9578009     |
| 2        | 583363  | 32.16705210 | 96.3573807    |
| 7        | 1559    | 0.08596437  | 0.2575089     |
| 9        | 2587    | 0.14264903  | 0.4273095     |
| NA       | 1208126 | 66.61692974 | NA            |

```

get_summary_stats_over_entire_period(analysis_data,MENTHLTH)

```

| MENTHLTH | n      | percent      | valid_percent |
|----------|--------|--------------|---------------|
| 1        | 61209  | 3.375108e+00 | 3.37514144    |
| 2        | 104960 | 5.787569e+00 | 5.78762674    |
| 3        | 67248  | 3.708103e+00 | 3.70813951    |
| 4        | 33564  | 1.850743e+00 | 1.85076128    |
| 5        | 89370  | 4.927926e+00 | 4.92797449    |
| 6        | 9240   | 5.095002e-01 | 0.50950525    |
| 7        | 35010  | 1.930476e+00 | 1.93049554    |
| 8        | 7119   | 3.925467e-01 | 0.39255064    |
| 9        | 1148   | 6.330154e-02 | 0.06330217    |
| 10       | 69186  | 3.814965e+00 | 3.81500328    |
| 11       | 505    | 2.784606e-02 | 0.02784634    |
| 12       | 4846   | 2.672119e-01 | 0.26721455    |
| 13       | 661    | 3.644801e-02 | 0.03644837    |
| 14       | 13450  | 7.416426e-01 | 0.74164996    |
| 15       | 65081  | 3.588613e+00 | 3.58864840    |
| 16       | 1131   | 6.236415e-02 | 0.06236477    |
| 17       | 921    | 5.078460e-02 | 0.05078510    |
| 18       | 1286   | 7.091096e-02 | 0.07091166    |
| 19       | 191    | 1.053188e-02 | 0.01053198    |

|    |         |              |             |
|----|---------|--------------|-------------|
| 20 | 40542   | 2.235515e+00 | 2.23553700  |
| 21 | 2363    | 1.302975e-01 | 0.13029880  |
| 22 | 606     | 3.341527e-02 | 0.03341560  |
| 23 | 432     | 2.382079e-02 | 0.02382102  |
| 24 | 443     | 2.442734e-02 | 0.02442758  |
| 25 | 14112   | 7.781458e-01 | 0.77815347  |
| 26 | 448     | 2.470304e-02 | 0.02470328  |
| 27 | 926     | 5.106030e-02 | 0.05106081  |
| 28 | 3801    | 2.095899e-01 | 0.20959193  |
| 29 | 1770    | 9.759906e-02 | 0.09760003  |
| 30 | 119032  | 6.563509e+00 | 6.56357457  |
| 77 | 25708   | 1.417557e+00 | 1.41757153  |
| 88 | 1028174 | 5.669425e+01 | 56.69481077 |
| 99 | 9041    | 4.985272e-01 | 0.49853214  |
| NA | 18      | 9.925328e-04 | NA          |

```
get_summary_stats_over_entire_period(analysis_data,EMPLOY)
```

| EMPLOY | n      | percent    | valid_percent |
|--------|--------|------------|---------------|
| 1      | 686415 | 37.8494129 | 38.136773     |
| 2      | 115136 | 6.3486812  | 6.396882      |
| 3      | 35525  | 1.9588739  | 1.973746      |
| 4      | 39761  | 2.1924499  | 2.209095      |
| 5      | 137597 | 7.5871968  | 7.644800      |
| 6      | 43732  | 2.4114137  | 2.429722      |
| 7      | 595120 | 32.8153415 | 33.064482     |
| 8      | 128512 | 7.0862434  | 7.140043      |
| 9      | 18079  | 0.9968890  | 1.004458      |
| NA     | 13665  | 0.7534979  | NA            |

```
get_summary_stats_over_entire_period(analysis_data,INCOME)
```

| INCOME | n      | percent   | valid_percent |
|--------|--------|-----------|---------------|
| 1      | 116808 | 6.440876  | 6.440876      |
| 2      | 191950 | 10.584260 | 10.584260     |
| 3      | 172331 | 9.502454  | 9.502454      |
| 4      | 200520 | 11.056816 | 11.056816     |
| 5      | 519465 | 28.643671 | 28.643671     |
| 6      | 167483 | 9.235132  | 9.235132      |
| 7      | 50144  | 2.764976  | 2.764976      |
| 9      | 394841 | 21.771814 | 21.771814     |

```
get_summary_stats_over_entire_period(analysis_data,EDUCA)
```

| EDUCA | n      | percent      | valid_percent |
|-------|--------|--------------|---------------|
| 1     | 2415   | 0.133164823  | 0.1331680     |
| 2     | 35024  | 1.931248353  | 1.9312941     |
| 3     | 71445  | 3.939528282  | 3.9396217     |
| 4     | 447823 | 24.693279781 | 24.6938653    |
| 5     | 515244 | 28.410921831 | 28.4115955    |
| 6     | 733517 | 40.446650808 | 40.4476098    |
| 9     | 8031   | 0.442835071  | 0.4428456     |
| NA    | 43     | 0.002371051  | NA            |

```
get_summary_stats_over_entire_period(analysis_data,INSURANCE)
```

| INSURANCE | n      | percent   | valid_percent |
|-----------|--------|-----------|---------------|
| 1         | 965814 | 53.255673 | 53.255673     |
| 2         | 90496  | 4.990014  | 4.990014      |
| 9         | 757232 | 41.754313 | 41.754313     |

Around 66% of women report missing responses when asked if they are currently pregnant.  
32% of women report they aren't currently pregnant.